STATS 201B Take-Home Final

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Due date: Monday, March 16, 2025 at 11:59 am on Gradescope

Please choose **one topic** from the following list and write an essay with no more than three pages using this LaTeX template available on Overleaf: `https://www.overleaf.com/latex/templates/fred-slash-uf-working-paper-template/mjnnjxzqmydm`.

Please attach your code to the report.

Note: Codes, references, supplementary proofs, and supplementary numerical results are not counted to three-page limit.

Topics

1. Double machine learning (DML) provides a framework for estimating treatment effects in high-dimensional settings by using machine learning methods to estimate nuisance functions while preserving valid statistical inference for the target parameter. Consider the partially linear model
$$Y = D\theta_0 + g_0(X) + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid X, D] = 0,$$
where D is the treatment variable, X is a vector of covariates, and θ_0 is the parameter of interest.

   (a) Design a simulation study to compare the performance of the DML estimator with ordinary least squares (OLS). Investigate how the choice of machine learning method used to estimate the nuisance function $g_0(X)$ (e.g., Lasso, random forests, or neural networks) affects the estimation accuracy of θ_0.

   (b) Propose and study extensions of the DML framework for estimating a more general treatment effect function $f_0(D)$ when the treatment effect is not linear in D, i.e., when the model is no longer of the form $f_0(D) = D\theta_0$.

2. The fixed-X knockoff framework provides a method for variable selection in linear models while controlling the false discovery rate (FDR). It treats the design matrix as deterministic and constructs knockoff variables that mimic the correlation structure of the original covariates. Consider the linear model
$$Y = X\beta_0 + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2),$$
where $X \in \mathbb{R}^{n\times p}$ is a fixed design matrix and $\beta_0 \in \mathbb{R}^p$ is a sparse regression coefficient vector.

   (a) Design a simulation study for linear regression with highly correlated covariates X. Consider several correlation structures for the design matrix X, such as autoregressive correlation and block correlation. Investigate how strong correlation among covariates affects the property of the knockoff method, in terms of false discovery rate and statistical power.

(b) Propose and study possible improvements or modifications to the fixed-X knockoff procedure for settings with highly correlated covariates. For example, you may consider grouping correlated variables or designing alternative feature importance statistics.

3. Conformal prediction provides a general framework for constructing prediction sets with marginal coverage guarantees under minimal assumptions. However, standard conformal prediction methods typically guarantee only *marginal* coverage, averaged over the covariate distribution, and do not in general provide *conditional* coverage given the feature value. Consider a regression setting with data

$$(Y_i, X_i), \quad i = 1, \ldots, n,$$

where $X_i \in \mathbb{R}^p$ is a vector of covariates and $Y_i \in \mathbb{R}$ is the response. Let $\widehat{C}(X_{n+1})$ denote a conformal prediction interval for a new response Y_{n+1} at feature value X_{n+1}.

(a) Design a simulation study to compare marginal coverage and conditional coverage under different data-generating mechanisms. For example, consider settings with heteroscedastic noise and nonlinear regression functions. Evaluate the performance of standard conformal intervals in terms of overall coverage, coverage across subgroups or local regions of the feature space, and average interval length.

(b) Propose and study modifications or extensions of conformal prediction that improve conditional validity or adaptivity, such as covariate-dependent score functions, or locally weighted conformity scores.

References

[1] Yoonkyung Lee, Yi Lin, and Grace Wahba. Multicategory support vector machines: Theory and application to the classification of microarray data and satellite radiance data. *Journal of the American Statistical Association*, 99(465):67–81, 2004.